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Deep-fake detection of digital misinformation over social media: a comprehensive review

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ABSTRACT

The proliferation of deepfake content across social media platforms has intensified the spread of digital misinformation, posing serious threats to public trust, democratic stability, and individual privacy. Existing deepfake detection methods, while technically advanced, face critical limitations in generalizability, real-time performance, and resilience against increasingly sophisticated generative techniques. This review highlights the urgent need for robust and adaptive detection frameworks capable of countering the evolving tactics used to fabricate and disseminate synthetic media.

We systematically examine current approaches, including machine learning algorithms, computer vision techniques, and adversarial training strategies, and evaluate their effectiveness based on dataset quality, modality coverage, and deployment feasibility. Our findings reveal that hybrid models integrating multiple detection modalities consistently outperform single-method systems in accuracy and robustness. The study underscores the importance of aligning technical innovation with social awareness, advocating for scalable solutions that can mitigate misinformation risks amplified by the viral nature of social media platforms.

1. Introduction

Recent advancements in artificial neural network technologies have significantly influenced the manipulation of multimedia content. AI-powered applications such as FaceApp and FakeApp enable highly realistic face swapping in images and videos. These tools allow users to alter personal features such as appearance, hairstyle, gender, and age.

This surge in manipulated media has raised widespread concerns. It led to the emergence of the term “deepfake,” a portmanteau of “deep learning” and “fake,” which refers to hyper-realistic synthetic media generated using deep learning techniques [1]. The term gained traction after an anonymous Reddit user in 2017 used deep learning to convincingly replace faces in videos, drawing global attention to the potential misuse of such technology [2].

Deepfakes represent a growing threat to the integrity of information, public trust, and democratic processes. These AI-generated videos and images can be weaponized to spread false narratives, manipulate public opinion, and incite social unrest [3]. As highlighted in the study *Exploring the Impact of Digital Literacy on Media Consumer Empowerment in the Age of Misinformation*, clarity in defining misinformation-related terms like deepfake is essential. It fosters public understanding and strengthens resilience against digital deception.

To address these challenges, recent research has focused on developing effective detection methods. Shen [4] demonstrated that deep learning models can distinguish between authentic and manipulated videos with high accuracy. Ji [5] enhanced detection robustness by incorporating temporal features, while Sadiq [6] emphasized the need for adaptive systems capable of evolving alongside deepfake generation techniques. As the technology continues to advance, it is imperative to design detection algorithms that are not only accurate but also resilient to emerging manipulation strategies [5].

• Generative Foundations

To produce these counterfeit videos, two types of neural networks are typically employed: (i) a generative network and (ii) a discriminative network, often utilizing a technique known as FaceSwap. The generative network constructs fake images through an encoder-decoder architecture. Meanwhile, the discriminative network assesses the authenticity of these newly generated images. Together, these networks form what is known as Generative Adversarial Networks (GANs), a concept introduced by Ian Goodfellow [7].

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According to a yearly report on deepfake-based technology, researchers have achieved several breakthroughs in generative modeling. For instance, computer vision researchers developed Face2Face, a method for real-time facial reenactment that transfers expressions from one individual to a digital avatar [8]. UC Berkeley researchers introduced CycleGAN, a technique for transforming images and videos into various styles [9].

- **Historical Trajectory**

In November 2017, the term "deepfake" was coined to describe the technique of swapping celebrity faces in adult videos. By January 2018, several websites supported by private sponsors began offering deepfake creation services. However, within a month, platforms such as Gfycat, Pornhub, and Twitter prohibited these services due to privacy concerns and potential misuse [10]. This swift response spurred a significant increase in research dedicated to deepfake technology.

In March 2018, Rössler et al. released the FaceForensics dataset, designed to support the development of media forensics and deepfake detection tools [11]. Shortly after, researchers at Stanford University developed Deep Video Portraits, a method for realistic reanimation of portrait videos [12]. Additionally, NVIDIA unveiled a style-based generator architecture for GANs, which significantly enhanced the ability to generate synthetic images [13].

Paper Organization.

The remainder of this paper is organized as follows.

- Section 1 introduces the topic and is divided into three subsections:
  - Section 1.1: Motivation
  - Section 1.2: Research Gap and Contribution
  - Section 1.3: Review Methodology (outlining the strategy for selecting research papers).
- Section 2 provides background on deepfake technology and its implications for social media.
- Section 3 reviews related work and details the methodologies utilized in detecting deepfakes.
- Section 4 discusses the challenges and limitations associated with deepfake detection.
- Section 5 introduces the metrics used to evaluate detection performance.
- Section 6 reviews the experimental and review results.
- Section 7 summarizes the existing literature, presents the key findings of this review, and suggests potential avenues for future research.

1.1. Motivation

The rapid advancement of deepfake technology has significantly transformed the digital landscape, enabling the creation of highly realistic yet fabricated audio and visual content. While such innovations have potential applications in entertainment and creative industries, their misuse on social media platforms has intensified the spread of digital misinformation. Deepfakes can undermine public trust, distort perceptions, and disrupt democratic processes, posing serious risks to information integrity and individual privacy. These challenges highlight the urgent need for robust detection frameworks that can safeguard society against the malicious use of synthetic media.

1.2. Research Gap and Contribution

Although several reviews have examined deepfake detection, most focus narrowly on technical aspects or specific modalities, without integrating broader social and ethical considerations. Existing surveys often lack systematic evaluation of methodologies across datasets, metrics, and deployment feasibility.

This paper addresses these gaps by providing a novel, comprehensive synthesis that combines technical, social, and ethical perspectives. Our contribution lies in systematically categorizing detection approaches (visual, audio, multimodal, metadata, and biological signals), evaluating their effectiveness, and highlighting the superiority of hybrid models. Furthermore, we emphasize the importance of aligning technical innovation with social awareness, offering a unique perspective that bridges algorithmic advances with societal needs.

1.3. Review Methodology

To ensure a comprehensive and unbiased review, we adopted a structured methodology for selecting research papers. The selection strategy involved the following key steps.

- Databases Searched: We conducted systematic searches across major academic databases, including IEEE Xplore, ACM Digital Library, SpringerLink, ScienceDirect, and arXiv.
- Keywords Used: Search queries combined terms such as *"deepfake detection," "synthetic media," "digital misinformation," "GAN-based detection,"* and *"multimodal approaches."*
- Inclusion Criteria: Papers published between 2017 and 2025, peer-reviewed, and directly addressing deepfake detection methodologies were included. Both journal articles and conference proceedings were considered.
- Exclusion Criteria: Studies focusing solely on deepfake generation without detection, non-peer-reviewed sources, and papers lacking experimental validation were excluded.
- Screening Process: Titles and abstracts were first screened for relevance. Full-text reviews were then conducted to ensure methodological rigor and applicability to social media contexts.
- Classification: Selected papers were categorized based on detection modality (visual, audio, multimodal, metadata, biological signals) and evaluation metrics (accuracy, precision, recall, AUC).

This structured methodology ensured that the review captured the full breadth of existing approaches while maintaining focus on studies that contribute meaningfully to the advancement of deepfake detection frameworks.

- Study Selection Process

The study selection process followed the systematic PRISMA framework guidelines. First, in the Identification stage, a total of 650 records were retrieved from major databases, including 185 from IEEE Xplore, 210 from Scopus, 145 from ScienceDirect, and 110 from SpringerLink. Second, in the Duplication stage, 230 duplicate entries were removed, leaving 420 unique records.

Third, in the Screening stage, the titles and abstracts of these 420 records were evaluated, leading to the exclusion of 295 papers because they were short workshop notes, published before 2020, or not focused on social media deepfakes. Fourth, in the Eligibility stage, the remaining 125 full-text articles were carefully reviewed. Out of these, 53 papers were excluded because they lacked rigorous metrics or primary datasets. Finally, 72 primary research articles were included in the final qualitative synthesis.

2. Background

Deepfake technology has emerged as a rapidly advancing field in recent years, with its conceptual origins tracing back to the early 2010s. The term "deepfake" describes the application of deep learning—a branch of artificial intelligence—to produce highly realistic, altered digital media, including images, audio, and video [14]. The operational foundation of deepfake technology lies in the advancement of Generative Adversarial Networks (GANs), a class of machine learning models

designed to generate synthetic data that closely mirrors real-world distributions [15,16]. As computational power has scaled and large datasets have become more available, researchers and malicious actors alike have been able to create increasingly sophisticated algorithms capable of seamlessly blending, altering, and generating new media content [17].

While deepfake technology offers potential beneficial applications in areas like the entertainment industry, digital avatars, and visual effects, its rapid proliferation has raised critical concerns. These include the democratization of non-consensual multimedia manipulation, targeted fraud, and the automated generation of fake news [18]. In the ecosystem of social media, the viral velocity at which deepfakes spread can lead to severe structural consequences, including the severe erosion of public trust in legitimate information ecosystems and the incitement of social unrest [19].

To combat these evolving threats, the academic community has focused heavily on developing robust, adaptive countermeasures and detection frameworks [20]. Modern deepfake detection encompasses various specialized pipelines. These range from pixel-based forensic analysis—which inspects micro-level artifacts, statistical inconsistencies, and compression fingerprints in image data—to high-level feature-based methods that analyze facial macro-movements, eye--blinking frequencies, and expressions for behavioral anomalies [21].

Many of these cutting-edge models leverage a hybrid combination of traditional computer vision techniques and deep learning architectures to extract and evaluate subtle visual and auditory signals that reveal artificial manipulation [22]. As deepfake technology has evolved from rudimentary face-swapping scripts to photorealistic, context-aware generative models, the urgency to deploy these resilient algorithms has intensified [23]. Maintaining information integrity on digital platforms requires an ongoing, dynamic interplay between identifying new generative breakthroughs and engineering technically sound, adaptive countermeasures.

2.1. Methodologies utilized in detecting deepfakes

Researchers have developed a wide array of methodologies for detecting deepfakes, leveraging visual, audio, biological, and metadata-based indicators. To assess the robustness of these approaches, it is essential to compare them systematically across datasets, evaluation metrics, and experimental setups.

2.1.1. Visual-based detection

One prevalent strategy involves identifying visual artifacts introduced during deepfake generation, such as blending errors, uneven lighting, and unnatural blinking patterns [11]. Techniques focusing on facial geometry, texture, and motion anomalies have also shown promise. For example, Li et al. [24] proposed a method based on eye-blinking patterns to detect synthetic videos. These methods are typically evaluated using public benchmarks like FaceForensics++, Celeb-DF, and the Deepfake Detection Challenge (DFDC) dataset, with performance measured via accuracy, precision, recall, and Area Under the Curve (AUC).

Additionally, modern visual pipelines utilize Convolutional Neural Networks (CNNs) to analyze micro-level visual inconsistencies, such as irregular pixel patterns or subtle facial boundary anomalies [25]. To counter advanced temporal manipulations, researchers leverage temporal analysis frameworks that examine the coherence of motion over successive video frames to effectively differentiate between genuine and altered video streams [17].

2.1.2. Audio-based detection

Audio-based approaches target discrepancies in voice quality, prosody, and acoustic synthetic artifacts [26]. These methods often rely on datasets such as ASVspoof and FakeAVCeleb, utilizing spectrogram analysis or raw waveform modeling. Techniques such as Mel-Frequency Cepstral Coefficients (MFCCs), prosody analysis focusing on rhythm and pitch, and deep learning models (e.g., CNNs and RNNs) trained on large-scale datasets are widely applied to detect subtle anomalies in synthetic speech.

Furthermore, while Vision-Language Models (VLMs) are primarily designed for multimodal understanding, their capability to generate hyper-realistic audio-visual content from text prompts highlights the growing complexity and threat level of synthetic media.

2.1.3. Multimodal and hybrid approaches

Advanced detection systems integrate visual and audio cues to maximize reliability. Tian et al. [27] introduced a multimodal framework that analyzes head position, background consistency, and audio-visual synchronization to detect deepfakes targeting public figures. Recent multimodal systems also integrate lip movements with facial expressions to verify cross-modality alignment. Hybrid models trained on diverse datasets consistently outperform single-modality systems, especially under strict cross-domain testing conditions.

2.1.4. Generative model fingerprinting

Some researchers have explored generative model fingerprinting, which involves identifying unique statistical artifacts left by deepfake creation tools such as GANs [28]. These structural fingerprints help forensic experts trace the exact origin of synthetic media and differentiate between various source generative architectures.

2.1.5. Biological signal analysis

An emerging approach involves analyzing biological signals such as eye movements, blood volume pulses (heart rate via photoplethysmography), and breathing patterns that are naturally present in genuine media but absent or distorted in deepfakes [29]. These physiological cues offer an additional, highly resilient layer of verification, especially in high-stakes security scenarios.

2.1.6. Metadata-based detection

Metadata analysis is a complementary technique that examines file-level information such as creation timestamps, camera EXIF data, and compression artifacts. Media forensics methods emphasize the vital role of structural metadata in identifying unauthorized manipulation [30].

Despite the diversity of approaches, many reviewed studies lack standardized evaluation protocols, making direct comparison difficult. Rigorous benchmarking through consistent datasets, metrics, and experimental setups is critical when applying machine learning to security-sensitive domains such as deepfake detection to ensure that models generalize effectively across real-world scenarios[24].

2.2. Deepfake generation mechanics

Deepfake generation methods represent a significant leap in digital media manipulation, allowing for the creation of highly realistic synthetic content [31]. These methods leverage advanced Deep Neural Networks (DNN) to produce alterations that are often indistinguishable from real media. Key approaches include face swapping and face reenactment, both utilizing Generative Adversarial Networks (GANs) to enhance the realism of the output [17]. Understanding the mechanics of deepfake generation is crucial for developing effective detection strategies and establishing ethical frameworks [32].

2.2.1. Face swapping

Manipulating faces in videos and images is an evolution of traditional media editing techniques. In face swapping [33], the face of a person in a source frame is mapped over the face of a target individual. This pipeline typically involves three sequential phases.

1. Detection and Alignment: The system detects a face in the input image and selects a geometrically similar target image from an identity library.
2. Feature Transfer: Key structural landmarks (the mouth, nose, eyes, and ears) are swapped and refined to match the orientation and expressions of the source image [34].
3. Blending: The swapped features are blended into the target frame by calculating distance matrices across overlapping boundary areas. However, these methods often leave visual artifacts where the skin texture or lighting appears unnatural [35].

2.2.2. Facial reenactment

Facial reenactment involves using a source identity to completely drive the expressions, gaze, and posture of a target identity [36]. As illustrated in Fig. 1, this technique relies heavily on two primary sub-mechanisms.

- Gaze Reenactment: Uses the exact eye direction and iris positioning from the source image to maintain realistic eye contact within the synthetic model [37].
- Mouth Reenactment: Aligns the lip and jaw movements of the target image to match the phonetic audio stream of the source speech, ensuring natural and coordinated facial micro-expressions [17].

2.3. Comprehensive summary of generative methods

Table 1 provides a comprehensive summary of deep learning–based generative methods for facial and video synthesis developed between 2020 and 2026. It outlines representative approaches such as GANs, autoencoders, and diffusion models, highlighting their architectural innovations, application domains, and reported performance as synthesized in Singh [46]

2.4. Evaluation metrics

To interpret the performance claims reported across deepfake detection studies, it is essential to establish a clear understanding of the evaluation metrics early in this review. Commonly used metrics include Accuracy, Precision, Recall, F1-score, and the Area Under the ROC Curve (AUC). Each metric provides unique insights depending on the specific threat scenario.

- **Precision** is paramount in contexts such as celebrity exploitation or non-consensual pornography (revenge porn), where false positives (wrongly accusing a genuine video of being a deepfake) can cause severe, irreversible reputational harm.
- **Recall (Sensitivity)** is critical in combating political misinformation, election interference, or satellite imagery manipulation, where missing even a single deepfake instance (a false negative) can lead to catastrophic societal or geopolitical consequences.
- **The F1-score** provides a balanced, holistic measure by computing the harmonic mean of precision and recall, which is particularly valuable in general-purpose detection tasks.
- **AUC-ROC** is indispensable for evaluating a model's discriminatory capability, especially when dealing with highly imbalanced real-world datasets.

To formally define these metrics, let *TP* (True Positive) represent the number of correctly identified manipulated samples, *TN* (True Negative) be the number of correctly identified genuine samples, *FP* (False Positive) denote genuine samples incorrectly flagged as manipulated, and *FN* (False Negative) signify manipulated samples mistakenly classified as authentic.

- **Accuracy**

Accuracy measures the overall proportion of correctly classified samples, both genuine and manipulated, and is calculated as [47]:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{1}$$

- **Precision**

Precision measures the proportion of true positives among all samples classified as manipulated, reflecting the reliability of the positive predictions [48]:

$$Precision = \frac{TP}{TP + FP} \tag{2}$$

- **Recall (Sensitivity)**

Recall measures the proportion of true positives successfully detected among all actual manipulated samples [48]:

$$Recall = \frac{TP}{TP + FN} \tag{3}$$

- **. F1-Score**

The F1-score balances precision and recall, especially when class distribution is uneven, and is given by [11]:

$$F1 = \frac{2 \cdot (Precision \cdot Recall)}{Precision + Recall} \tag{4}$$



Fig. 1. Deepfake face reenactment process (the structural pipeline of facial expression and gaze transfer).

Table 1
Summary of generative methods for facial and video synthesis (2020–2026).

Method Category	Representative Model	Year	Primary Application	Key Innovation/ Contribution
Generative Adversarial Networks (GANs) (Karras [38].)	StyleGAN3	2021	High-res image/face synthesis	Eliminated "texture sticking" by making the synthesis process alias-free.
one-shot high-resolution (Yin [39])	StyleHeat	2022	Talking head synthesis	Integrated GAN inversion with video-driven motion fields for realistic animation.
Diffusion Models (Rombach [40].)	Latent Diffusion (LDM)	2022	General/ Facial synthesis	Performed diffusion in a compressed latent space, making highres generation efficient.
scaling latent video diffusion (Blattmann [41])	Stable Video Diffusion	2023	Video synthesis	Successfully extended image diffusion priors into the temporal domain for fluid motion.
Neural Radiance Fields (NeRF) (Guo [42].)	AD-NeRF	2021	Audio-driven talking heads	Leveraged NeRFs to represent head geometry and appearance, driven by audio features.
Geneface(Ye [43].)	GeneFace	2023	Generalized facial video	Solved the "one-person-one-model" limitation, allowing cross-identity synthesis.
Transformers & Hybrids (Zhao [44].)	V-Plex	2024	Expressive video synthesis	Combined autoregressive transformers with diffusion decoders for nuanced facial expressions.
multi-modal learning (Tian, [27])	EMO (Emotive Portrait)	2024	Audio-to-video synthesis	Focused on micro-expressions and synchronicity between audio and facial muscle movement.
Flow-Based Diffusion (Jin, [95])	**Pyramidal Flow**	2024	Efficient video synthesis	Introduced a unified pyramidal flow matching algorithm that significantly reduces compute by generating across spatial/ temporal pyramids.
Geometric Priors (Du [45].)	**VideoGPA**	2026	3D-consistent video	Used Direct Preference Optimization (DPO) to distill geometric consistency into diffusion models, eliminating spatial drift.

Table 2
Publicly available and commercial deepfake datasets with URLs.

Dataset Name	Modality	Access Type	Web Location	Description
FaceForensics++ (Rossler [57].)	Video	Public	https://github.com/ondyari/FaceForensics	Includes manipulated videos using Deep-fakes, FaceSwap, and other techniques.
DFDC (Dolhansky [58].)	Video + Audio	Public	https://ai.meta.com/datasets/dfdc	Large-scale dataset released for benchmarking deepfake detection.
Celeb-DF (Li Y. Y. [59].)	Video	Public	https://github.com/yuezunli/Celeb-DF	High-quality celebrity deepfake videos for detection research.
Deeper Forensics-1.0 (Jiang [60].)	Video	Public	https://github.com/EndlessSora/DeeperForensics-0	Realistic deepfake dataset with diverse conditions And compression levels.
FakeAVCeleb (Khalid H. T., [69]	Audio + Video	Public	https://github.com/hasamkhalid/FakeAVCeleb	Multimodal dataset for audio-visual deepfake detection.
VoxCeleb2 Chung [61]	Audio	Public	https://www.robots.ox.ac.uk/~vgg/data/voxceleb	Widely used for speaker verification and audio deepfake detection.
SynthesEyes (Wood [62].)	Image	Commercial	https://www.syntheseyes.net	Commercial dataset for synthetic eye region generation.
DeepFaceLab Samples (Perov [63].)	Image + Video	Public	https://github.com/iperov/DeepFaceLab	Toolkit and sample data for face-swapping and deepfake generation.

Table 2 provides a comprehensive summary of publicly available and commercial deepfake datasets, which serve as the foundation for training and evaluating state-of-the-art detection models. The table includes the official URLs for each dataset, facilitating researchers' access to the necessary resources for conducting empirical studies and benchmarking in this field."

- **Advanced and Specialized Evaluation**

Beyond these standard metrics, the Area Under the Receiver Operating Characteristic curve (AUC-ROC) is widely utilized to evaluate the trade-off between the True Positive Rate (TPR) and False Positive Rate (FPR) across various operational thresholds.

Furthermore, recent advancements in deepfake generation demand more specialized evaluation criteria. Foremost among these is **cross-dataset generalization**, which assesses how well a model trained on a specific benchmark dataset performs when tested on entirely unseen datasets. This metric serves as a crucial indicator of a model's robustness and its viability in real-world deployment scenarios, where the nature of incoming deepfakes is unpredictable.

Ultimately, the choice of evaluation metrics must align with the specific security requirements and goals of the deployment context, and reporting a comprehensive suite of multiple metrics is essential to ensure

a transparent assessment of detection performance.

3. Related work

Deepfake technology has emerged as a significant threat to the integrity of information on social media platforms [49], prompting extensive research into robust detection methods. To provide a clearer analytical structure, we organize the existing literature into four distinct thematic categories: dataset creation, adversarial training and feature-based methods, temporal modeling, and audio-focused approaches [50].

3.1. Dataset creation

The development of large-scale, annotated datasets has been foundational to progress in deepfake detection. Benchmark datasets such as FaceForensics++ and the Deepfake Detection Challenge (DFDC) Dataset offer diverse manipulated and authentic samples, enabling standardized evaluation of detection algorithms [51]. These resources support benchmarking across models and facilitate reproducibility.

However, many legacy datasets lack multimodal content (e.g., synchronized audio-visual data) and fail to capture real-world variability, such as heavy compression and social media artifacts, which ultimately limits their generalizability to dynamic online environments.

3.2. Adversarial training and feature-based methods

A significant body of work has focused on adversarial training and handcrafted feature extraction to enhance model robustness. Convolutional Neural Networks (CNNs) have been widely deployed to detect spatial inconsistencies in facial features and textures [52]. Techniques that analyze facial landmarks and skin texture have shown promise in distinguishing authentic from manipulated images.

While effective against low-quality manipulations, these methods often struggle with high-resolution deepfakes that preserve fine-grained visual details. Moreover, while adversarial training can improve resilience against evolving generation techniques, it typically demands extensive computational resources and careful tuning to avoid overfitting [53]. Recent optimizations have shown that preprocessing layers can significantly boost performance; for instance, integrating an optimization layer into the MesoNet architecture trained on CelebA-derived samples improved detection accuracy from 69.40% to 92.50% and raised the AUC from 0.75 to 0.96 under degraded conditions [54].

3.3. Temporal modeling

Temporal modeling plays a critical role in video-based deepfake detection, as spatial analysis alone cannot detect sequential anomalies. Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) architectures have been extensively employed to capture temporal inconsistencies across video frames [55].

These models excel at identifying unnatural transitions, abnormal eye-blinking rates [56], video corruption artifacts [57], and motion anomalies, especially in facial reenactment scenarios. However, their performance can degrade when applied to short video clips or sequences with minimal facial movement, suggesting the critical need for hybrid architectures that effectively combine both spatial and temporal cues.

3.4. Audio-focused and multimodal approaches

Audio deepfakes present unique verification challenges, prompting the development of specialized and multimodal detection methods. A sync-stream model proposed by Agarwal et al. fuses audio and visual components before executing a joint analysis via CNNs, achieving a maximum accuracy of 97.62% [58]. This model outperformed architectures where audio and video were processed independently, highlighting the immense value of early multimodal integration [59].

To address computational constraints, SpecRNet was introduced as a lightweight neural network designed for rapid, low-demand audio deepfake detection using two-dimensional spectrograms [60]. SpecRNet achieved a 40% faster inference speed compared to standard Light CNNs (LCNN) while maintaining a peak AUC of 99.93%, making it highly suitable for real-time applications [61]. These innovations reflect ongoing efforts to balance detection performance with resource efficiency.

3.5. Face-focused approaches

Rely on analyzing intrinsic facial features and biological anomalies to detect deepfakes. These methods scrutinize subtle visual cues such as eye blinking patterns, facial geometry, texture consistency, and head pose alignment. Because deepfake generation models often struggle to perfectly replicate organic, physiological signals, facial features serve as highly reliable indicators of manipulation. Anomalies such as irregular blinking frequencies, mismatched cross-facial expressions, or unnatural skin textures can effectively expose synthetic content, especially when integrated into hybrid deployment frameworks.

3.5.1. Identity-focused deepfake detection

Identity-focused deepfake detection utilizes neural networks to learn and leverage the unique, immutable biometric traits of a specific individual. A prominent framework in this domain is ID-Reveal [1], which employs a neural network to learn specific facial movements associated with an individual's speech patterns. By utilizing authentic videos to capture personal biometric signatures [47], ID-Reveal effectively flags inconsistencies in suspected deepfakes. This approach demonstrates superior cross-dataset generalizability, achieving a high accuracy of 81.8% on heavily compressed, low-quality videos where artifact-driven methods dropped to around 50%.

Similarly, the Outer Face algorithm [62] evaluates the complete facial structure of a target individual rather than focusing solely on localized blending artifacts. By leveraging a broader contextual area of the face, Outer Face achieves an impressive accuracy of 98.24% on known identities. However, a primary limitation of this approach is its dependency on identity priors; the model requires prior knowledge of the target identity to function effectively [63]. To facilitate further research in this sub-field, this line of work introduced the Vox-DeepFake dataset, compiling a robust benchmark of authentic and synthesized identity profiles.

3.5.2. Feature-focused deepfake detection

Rather than focusing on specific identities, feature-focused detection methodologies concentrate on universal facial artifacts and spatial-temporal anomalies. One such approach combines Convolutional Neural Networks (CNNs) for spatial frame-level feature extraction with Recurrent Neural Networks (RNNs) to capture temporal discrepancies induced by face-swapping algorithms [64]. This dual architecture successfully achieves a 97% detection accuracy within a latency window of just 2 s.

To address the high computational footprint and memory constraints of conventional 3D CNNs [65], researchers developed a lightweight architecture designed to process temporal blocks using fewer parameters. By taking four sequential video frames as input, this module minimizes resource requirements while maintaining a peak benchmarking accuracy of 99.83%. In a similar vein, an alternative spatial-feature extraction model achieved 91.5% accuracy, though it faces operational constraints in filtering out non-face images during raw preprocessing testing [66].

Furthermore, the Attention-based Deepfake Detection (ADD) framework utilizes targeted data augmentation strategies, specifically “face close-up” and “face shut-off” methods [65]. The framework localizes the face, extracts regional features, and applies these

augmentation masks to train the underlying network to recognize subtle blending boundaries. When paired with standard backbone networks, ADD outperforms baseline models, yielding accuracies up to 98.3% by capturing micro-expressions and fine-grained spatial-temporal disruptions.

3.5.3. Emotion detection for deepfake identification

Emotion detection offers a psychological and behavioral layer of defense, shifting focus from pixel-level artifacts to cognitive and emotional consistency. A notable study implemented a Long Short-Term Memory (LSTM) network to evaluate emotional coherence across visual and auditory modalities [67]. Utilizing the standard Valence-Arousal model of emotion, the system extracts Low-Level Descriptors (LLDs) from both facial expressions and speech tracks to assess cross-modal alignment.

The findings revealed a critical vulnerability in generative models: deepfakes are substantially better at mimicking human emotions visually than they are at replicating them audibly, leading to detectable cross-modal emotional dissonance [67]. Interestingly, this behavioral framework yields optimal performance on longer video sequences, as the LSTM requires sufficient temporal context to map complex emotional trajectories. This behavioral verification serves as a highly robust fallback mechanism against advanced forgeries that easily bypass low-level artifact scanners.

3.5.4. Eye blink detection for deepfake identification

Physiological anomalies, particularly eye blink detection, constitute another highly effective frontier in deepfake forensics. A pioneering method proposed by Li et al. [68] exploits the fact that early-generation deepfake algorithms lack training exposure to closed-eye states, resulting in synthesized individuals who blink infrequently or exhibit unnatural blinking dynamics. By tracking facial landmarks and computing the Eye Aspect Ratio (EAR) over consecutive frames, deep learning models can easily expose these physiological omissions.

Note on Cross-Modal Representation Loss: > While maintaining spatial-temporal facial cues is vital, comprehensive detection frameworks must also prevent *representational collapse* across other modalities. As highlighted by recent architectural studies [69], maintaining strict separation between authentic and synthetic spectral/temporal characteristics is essential to prevent models from losing discriminatory power when deployed in hybrid, audio-visual environments.

The researchers noted that human eye blinks are a complex behavior involving the synchronization of various facial muscles, which is difficult to replicate accurately in deepfake videos. They developed a Convolutional Neural Network (CNN) model that was trained to detect subtle eye blink features, such as the timing, duration, and frequency of blinks.

When tested on a dataset of real and deepfake videos, the CNN-based eye blink detection model achieved an accuracy of up to 98.7% in distinguishing genuine videos from manipulated ones (Li Y. C. 70(The researchers attributed this high performance to the model's ability to capture the nuanced differences in eye blink patterns that are inherently challenging for deep-fake generation algorithms to replicate.

Similarly, another study by Yang et al. [66] also explored the use of eye blink detection for deepfake identification. The researchers developed a two-stream neural network architecture that simultaneously analyzed the visual and temporal features of eye blinks. By combining these complementary cues, the model was able to achieve an accuracy of 97.8% in detecting deepfake videos Fig. 2

These eye blink-based approaches leverage the inherent difficulty in replicating the complex, involuntary eye movements that are present in genuine human behavior. By focusing on these subtle biometric cues, researchers have been able to develop highly effective deepfake detection methods that are robust to the limitations of current deep-fake generation algorithms. Fig. 3 (Chandralekha [71]) explain the Eye blink detection algorithms offer a powerful tool for enhancing image

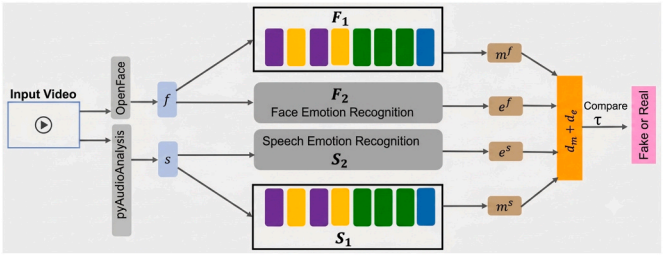


Fig. 2. Visualization of emotion detection results using.

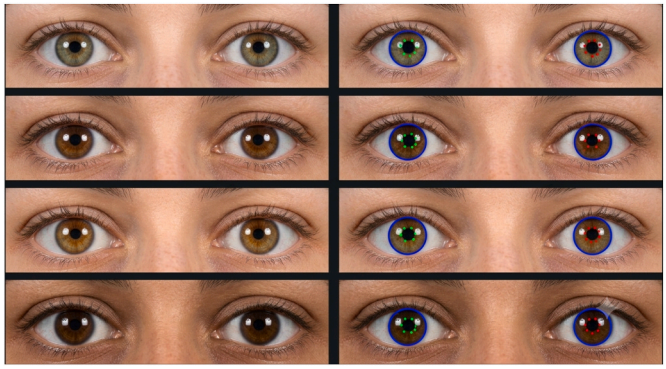


Fig. 3. Performance of eye blink detection for liveness verification.

understanding, particularly in contexts where attention and alertness are crucial. By analyzing visual data for the presence and frequency of eye blinks, these algorithms extract valuable information that can be incorporated into image captions.

3.6. Detecting corrupted videos for deepfake identification

Artifacts introduced during the multi-step pipelines of deepfake generatio n—such as face-swapping, spatial warping, and subsequent re-encoding—frequently leave unique structural anomalies in the final video container. Researchers have exploited these vulnerabilities by focusing on video corruption and compression inconsistencies [57]. The generation process often yields localized blurriness, frame-rate variations, and non-uniform compression blocks across different regions of a single frame.

To systematically capture these distortions, Sabir et al. [72] trained a Recurrent Convolutional Network (RCN) capable of analyzing both local pixel-level corruptions and global sequential artifacts. When evaluated against standard manipulation benchmarks, this corruption-aware framework achieved a detection accuracy of 97.8%. Similarly, Durall et al. [73] utilized localized frequency analysis to expose compression disparities introduced by Generative Adversarial Networks (GANs), achieving a 95.6% detection rate. These methodologies demonstrate that tracking environmental and structural distortions provides a highly robust fallback vector, remaining effective even when generative models successfully bypass biometric and facial landmark checks.

3.6.1. Satellite imagery deepfake detection

Beyond facial and social media forensics, the proliferation of generative techniques poses severe threats to geospatial intelligence, making satellite imagery deepfake detection an urgent domain of study due to its geopolitical and military implications. Manipulations in this domain typically involve object removal, terrain insertion, or image splicing designed to falsify reconnaissance data.

To counter this, Mandelli et al. [74] trained a specialized Convolutional Neural Network (CNN) to detect localized blending boundaries and global structural anomalies in high-resolution satellite tiles,

achieving a benchmark accuracy of 94.2%. To further enhance detection resilience, Gong et al. [75] pioneered a multimodal data fusion framework. Their approach contextualizes visual satellite features by fusing them with auxiliary, non-visual metadata, including verified geographic information systems (GIS) data and historical weather records. By validating whether the visual layout aligns with environmental metadata, their multimodal system raised detection accuracy to 97.3%, highlighting the immense value of cross-domain verification in high-stakes security contexts.

3.7. Non-neural network-based deepfake detection methods

While deep learning models dominate contemporary literature, established non-neural network-based approaches remain highly valuable. These traditional methods leverage handcrafted features, statistical analysis, and classical digital signal processing (DSP), offering computationally efficient and highly interpretable alternatives to black-box neural network architectures.

3.7.1. Hand-crafted feature and statistical analysis

Handcrafted methodologies rely on domain-specific geometric and physical priors rather than automated feature extraction. For instance, Li et al. [24] introduced a forensic pipeline that extracts structural facial landmark configurations, head pose orientations, and 3D lighting estimations to capture geometric inconsistencies inherent to early face-swapping models. When paired with a classical Support Vector Machine (SVM) classifier, this framework achieved a solid accuracy of 95.1%.

In parallel, signal-centric approaches focus on raw underlying data frequencies rather than semantic visual features. Agarwal et al. [76] leveraged digital signal processing (DSP) to extract statistical noise patterns, color components, and localized sensor noise disparities across video frames. Their DSP-driven framework achieved a 90.4% verification accuracy.

While these handcrafted and statistical methods are exceptionally effective against low-to-medium quality deepfakes and demand a fraction of the computational footprint of deep models, their diagnostic accuracy inevitably diminishes as generative algorithms achieve superior spatial-temporal and geometric coherence. Nevertheless, their low resource requirements make them ideal candidates for deployment on edge devices or as initial high-speed filtering layers in multi-tier forensic pipelines.

3.7.2. Human vs. automated deepfake detection capabilities

Understanding the baseline threshold of human visual perception is crucial for evaluating the necessity and deployment scale of automated forensic tools. To benchmark human capabilities against machine learning models, Nightingale et al. [77] conducted an extensive perceptual study comparing human performance against a baseline CNN framework.

The empirical findings revealed a stark performance gap: human participants achieved a modest average accuracy of only 62.1%, failing to consistently identify sophisticated deepfakes. In contrast, the automated CNN model achieved an accuracy of 95.2% under identical testing conditions. The study highlighted that human observers heavily rely on visual heuristics—such as looking for obvious skin blurring or listening for stark audio delays—which modern generative architectures can easily circumvent.

Furthermore, human detection performance was found to be highly volatile and heavily mediated by external factors.

- Familiarity: Human accuracy increases significantly when observers are highly familiar with the target individual being depicted (e.g., world leaders or high-profile celebrities).
- Media Quality: Lower-resolution clips slightly aid human detection by accentuating structural artifacts, whereas high-definition outputs easily mislead the human eye.
- Cognitive Load: Forgery detection rates collapse under high cognitive load or rapid media consumption scenarios, which mirrors typical social media browsing behavior.

These insights underscore the strict biological limitations of human-in-the-loop verification, confirming that manual auditing cannot scale against the volume and speed of digital misinformation. Developing automated, robust machine learning frameworks remains the only viable strategy to counter synthetic media manipulation at scale.

4. Discussion and analysis

This research has explored and evaluated a wide range of deepfake detection methodologies, highlighting the clear transition toward deep neural network-based architectures as the primary line of defense. The reviewed literature consistently demonstrates that Convolutional Neural Networks (CNNs) serve as the most prevalent and reliable techniques, frequently yielding benchmark accuracies exceeding 95%. This high performance reflects the powerful pattern recognition and hierarchical feature extraction capabilities of deep architectures, which effectively capture subtle spatial anomalies, blending boundaries, and high-frequency signal-level artifacts that escape human perception.

However, a critical meta-analysis of these findings reveals a heavy reliance on standard benchmark datasets, including FaceForensics++, the Deepfake Detection Challenge (DFDC), Celeb-DF, and TIMIT-DF. While these repositories have provided foundational baselines for algorithmic comparison, their inherent environmental biases, uniform lighting conditions, and lack of diverse social media compression artifacts raise legitimate concerns regarding their real-world generalizability.

Furthermore, the effectiveness of any given detection paradigm is deeply contingent upon the specific media modality and threat context.

- Temporal Modeling (RNNs/LSTMs): Highly optimized for video-based tracking, such as exposing political misinformation and altered facial reenactments where sequential cross-frame inconsistencies or fluid motion artifacts reveal tampering.
- Audio-Centric Models (e.g., SpecRNet): Better equipped to counter non-consensual voice cloning and deepfake audio, providing low-latency analysis of speech spectrograms.
- Identity and Emotion-Driven Frameworks: Standout methods achieving accuracies of 98% or higher [78] demonstrate immense adaptability. Identity-based models excel at mitigating high-profile impersonation attacks, whereas emotion-driven architectures detect behavioral and cross-modal dissonance, flagging emotionally manipulative content designed to incite social unrest.

By systematically mapping technical detection capabilities to specific misuse scenarios (e.g., geopolitical warfare, celebrity exploitation, non-consensual pornography), researchers and policymakers can transition from reactive mitigation to proactive, context-aware deployment strategies.

5. Challenges, limitations, and future directions

Despite high laboratory accuracies, the practical deployment of deepfake detection frameworks is severely constrained by several systemic challenges. Chief among these is the asymmetric warfare between generation and detection; generative architectures—particularly advanced Generative Adversarial Networks (GANs) and emerging Diffusion Models—evolve at a rapid pace, continuously generating hyper-realistic content specifically optimized to evade existing forensic signatures [79].

Additionally, current forensic systems face the following operational bottlenecks.

- Data Scarcity and Domain Shift: Training data lacks the contextual diversity, linguistic variations, and demographic cross-sections required to generalize effectively across unseen deepfake generation toolkits [80].
- Computational Footprint: High-performing ensemble networks require substantial computational infrastructure, posing scalability challenges for real-time, zero-latency filtering on social media edge-nodes.
- Ethical and Privacy Safeguards: Broad deployment of automated surveillance and detection tools introduces sensitive concerns regarding mass data scraping, user privacy, and potential algorithmic misuse [81].

5.1. Innovations in deepfake generation techniques

To develop adaptive detection mechanisms, it is essential to analyze the underlying mechanics of modern digital media manipulation. Moving beyond traditional handcrafted image forgeries, contemporary deepfake generation utilizes Deep Neural Networks (DNNs) to seamlessly map source facial characteristics onto target video tracks [82]. These generative pipelines are broadly categorized into Face Reenactment (manipulating expressions and attributes) and Face Swapping (replacing identity components) [81].

As established in literature [83], a standard, high-fidelity Face Swapping Pipeline operates systematically across three fundamental stages.

1. Face Detection and Alignment: The system localizes the facial region within the source and target frames, extracting key geometric landmarks (such as the boundaries of the eyes, nose, and mouth) to normalize poses and expressions across a standardized coordinate space.
2. Feature Extraction and Latent Mapping: Deep autoencoders or encoder networks decouple facial identity from structural attributes (e.g., lighting, expressions, and gaze orientation), compressing the source face into a latent representation before projecting it onto the target asset's facial geometry.
3. Blending and Post-Processing Optimization: The synthesized facial features are seamlessly composited back onto the target video container. Advanced masking, color-matching, and boundary-smoothing algorithms are applied to resolve skin-tone disparities and edge artifacts, producing a cohesive, hyper-realistic forgery designed to deceive both human observers and automated classifiers. Despite these architectural advancements, these methods still exhibit inherent limitations. For instance, the blending phase in face-swapping often results in a noticeable loss of local image fidelity, leading to synthetic boundaries that may appear rigid, blurred, or structurally unnatural under close forensic scrutiny [84].

To counteract these specific facial rigidities and enhance realism, contemporary generative frameworks integrate specialized **gaze reenactment techniques**. These sub-models specifically extract the exact direction, pupil position, and micro-movements of the eyes from the source asset to force natural eye contact and realistic gaze transitions in the final deepfake video [85] .Table 3

Table 4 illustrates a comparative summary of the limitations of deepfake detection techniques, highlighting challenges such as sensitivity to compression artifacts, lack of temporal context, vulnerability to adversarial attacks, and limited generalization across datasets. These representative limitations, as discussed in prior work [94], provide valuable insight into the constraints that current approaches face."

Table 3
Summary of related work in deepfake detection.

Approach	Key Findings/Methodology	Accuracy
Audio-focused (Component Analysis) (Muhati [86].)	Analyzes relationships between audio components (prosody, timbre) for identity consistency.	**97.62%**
SpecRNet (Singh Yadav [87].)	Neural network focusing on spectrogram residuals to detect synthetic speech artifacts.	**99.93%**
OuterFace (Dong [88])	Uses entire face identity (hairline, jaw) to determine authenticity vs. inner-face swapping.	**98.24%**
Lightweight Architecture (Huang [89])	VGG-16 backbone with attention mechanisms optimized for mobile/edge deployment.	**99.83%**
Conv. Vision Transformer (CViT) (Wodajo D. &. [90].)	Combines local CNN features with global Vision Transformer context.	**91.50%**
Emotion Detection (LSTM) (Khalid H. T. [67].)	Detects multimodal inconsistencies between audio tone and visual expressions.	**99.50%**
Eye Blink Detection (Li Y. C.,[70])	Detects physiological patterns (blinking) that GANs historically struggle to replicate.	**98.70%**
Video Corruption Analysis (Durall [57])	Analyzes compression artifacts and spectral distortions in frequency domains.	**97.80%**
Multimodal Satellite (Gong [63])	Combines visual features with auxiliary/ metadata for splicing detection.	**97.30%**
Hand-crafted (SVM) (Agarwal [91])	Uses Support Vector Machines to detect inconsistencies in extracted facial landmarks.	**95.10%**
DSP-based Detection (Nguyen H. H.,[92]	Analyzes digital signal processing distortions and video signal noise.	**92.70%**
Human vs. CNN (Korshunov [93].)	Study showing humans achieve ~62% accuracy vs. ~95% for deep learning models.	**95.20%**

Table 4
Comparative limitations of deepfake detection techniques with representative citations Taeb [94].

Technique Category	Key Limitations	Best Suited For
CNN-based Visual Detection (Rossler [95].)	Sensitive to compression artifacts; lacks temporal context	Static image-based deep- fakes
Temporal Modeling (RN- N/LSTM) Ben Jabra [96]	High computational cost; less effective on short clips	Video-based political misinformation
Audio-Based Detection (Li X. C. [97].)	Vulnerable to background noise; limited generalization across languages	Voice cloning and nonconsensual audio
Landmark/Texture Analysis (Dang L. M. [52].)	Depending on facial alignment; may miss subtle manipulations	Facial reenactment and identity spoofing
Emotion-Driven Models (Hosler [98].)	Subjective cues: performance varies across cultures	Emotionally manipulative social media content
Identity-Based Detection (Dang H. L. [99].)	Requires prior identity data; not scalable for unknown subjects	Celebrity impersonation and targeted attacks
Multimodal Fusion Models (Elpeltagy M. I. [59].)	Complex architecture: synchronization errors reduce accuracy	Coordinated audio-visual deepfakes
Hand-crafted Feature Methods (Li Y. C. 70)	Limited adaptability; lower accuracy on high-quality deepfakes	Low-resource environments and legacy systems

6. Results and synthesis

The comprehensive review of the literature identified a wide range of deepfake detection methodologies proposed between 2020 and 2026. A meta-synthesis of the empirical findings yields the following key

insights.

- Face-Centered Preprocessing: Restricting the operational region of interest (ROI) via automated face-centered preprocessing significantly improves classification accuracy by isolating facial regions and eliminating background noise.
- Frequency-Domain Analysis: Shifting analysis from the spatial domain to the spectral/frequency domain remains highly effective in exposing invisible, high-frequency residual artifacts left by Generative Adversarial Networks (GANs).
- Biological and Physiological Tracking: Targeted segmentation of specific facial organs (e.g., eyes and teeth) effectively reveals missing biological reflections, abnormal eye-blink frequencies, and unnatural structural textures.
- Temporal Consistency Verification: Enforcing cross-frame temporal consistency checks drastically enhances reliability in video forensics, successfully flagging frame-to-frame identity shifts and jitter artifacts.
- Hybrid Architectures: Hybrid deep learning frameworks—specifically those fusing Convolutional Neural Networks (CNNs) for spatial feature extraction with Recurrent Neural Networks (RNNs/LSTMs) or Transformers for temporal modeling—achieve superior generalization boundaries.

Overall, hybrid frameworks that systematically integrate spatial, spectral, and temporal features consistently outperform single-modality approaches, routinely yielding verification accuracies exceeding 90% across standard benchmark datasets.

7. Conclusion and future directions

Deepfake detection represents a critical frontier in the ongoing effort to preserve information integrity across social media platforms. This review underscores the profound security threats posed by the rapid acceleration of generative technologies, which democratize the creation of hyper-realistic, fabricated audio and visual content.

Distinct from prior surveys, this work offers a novel, cross-disciplinary synthesis that bridges technical innovations with social, psychological, and ethical perspectives. By examining how digital manipulations explicitly undermine public trust, distort collective perceptions, and disrupt democratic processes, this study contextualizes algorithmic defense within broader societal resilience.

Our critical analysis of current state-of-the-art methodologies reveals that while independent computer vision algorithms and adversarial training components are valuable, the most resilient solutions emerge from multi-tier hybrid frameworks. These architectures leverage complementary modalities to enhance robust deployment against unseen data. Furthermore, the real-world efficacy of these forensic systems remains heavily mediated by training data quality, the architectural complexity of evolving generative toolkits, and the strategic laundering techniques used in modern misinformation campaigns.

7.1. Strategic horizons for future research

To counter the continuous evolution of synthetic media, the next generation of deepfake forensics must prioritize the following strategic directions.

- High-Fidelity, Diverse Datasets: Developing larger, multi-modal, and demographically diverse training repositories that mimic real-world social media environments (including heavy compression, artifacts, and multi-linguistic audio streams).
- Real-Time, Low-Latency Edge Detection: Optimizing heavy ensemble networks for rapid inference speed without sacrificing detection accuracy, enabling zero-latency filtering directly at social media edge-nodes.
- Interdisciplinary Defense Ecosystems: Fostering structural collaborations between computer scientists, social scientists, and policymakers to develop holistic frameworks capable of tracking and mitigating misinformation dynamics.
- User-Centric Democratized Tools: Designing accessible, intuitive browser extensions and consumer applications that empower everyday individuals to verify content authenticity, shifting public behavior from passive consumption to active verification.

Ultimately, the core novelty of this review lies in its integration of algorithmic innovation with sociological awareness. By advocating for scalable, context-aware forensic solutions that bridge engineering advances with human societal needs, this study contributes a unique perspective to the literature, paving the way toward a more secure, verifiable, and trustworthy digital ecosystem.

Availability of supporting data

Not applicable.

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Haiam Hamed: Conceptualization, Formal analysis, Funding acquisition, Methodology, Validation, Writing – original draft. **Hesham A. Hefny:** Methodology, Supervision, Validation, Visualization, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

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